# SIDDHARTH SINGH

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AI Engineer with 5+ years of experience designing and developing intelligent software solutions across Generative AI, Agentic AI, semantic search, and backend platform engineering. Skilled in building enterprise-scale RAG systems, document intelligence platforms, REST APIs, and knowledge retrieval solutions using FastAPI, PostgreSQL, vector search, and modern AI technologies. Passionate about translating complex business challenges into scalable and user-centric products.

## Technical Skills

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| **AI Engineering & Retrieval** | Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Search, Information Retrieval, Prompt Engineering, Embeddings, LLM Applications |
| **Backend & Platform Engineering** | FastAPI, Django, Django REST Framework, SQLAlchemy, REST APIs, Microservices, System Design |
| **Database & Search** | PostgreSQL, LanceDB, Database Design, Metadata Filtering |
| **AI Frameworks & Tools** | OpenAI SDK, LangChain, LangGraph, CrewAI, Chainlit, Gradio |
| **Programming** | Python, SQL |
| **Security & Authentication** | Keycloak, RBAC |
| **Data Analytics** | Power BI, Tableau, Microsoft Excel |

## Experience

May 2024 - PRESENT

**TATA CONSULTANCY SERVICES, Bengaluru** – *Researcher*

- Designed and developed enterprise AI solutions focused on intelligent search, knowledge retrieval, and AI-powered document interactions using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).
- Contributed to the development of a custom Semantic Knowledge Engine (SKE), implementing chunking strategies, embedding-based indexing, vector retrieval, metadata-driven search, and retrieval quality improvements for enterprise AI applications.
- Contributed to enterprise document intelligence initiatives involving parser orchestration, OCR-enabled extraction, hierarchy-aware content processing, and standardized document modeling to improve downstream retrieval accuracy and answer grounding.
- Developed and enhanced FastAPI-based backend services, REST APIs, database integrations, and platform capabilities supporting enterprise AI applications and knowledge platforms.
- Implemented semantic search and retrieval workflows leveraging embeddings, vector search, metadata filtering, and similarity-based ranking to improve answer relevance and user experience.
- Evaluated retrieval effectiveness and search quality using similarity scoring, ranking metrics, and comparative analysis methodologies to improve the relevance of enterprise knowledge retrieval systems.
- Contributed to platform modernization initiatives involving PostgreSQL search capabilities, vector database integrations, API enhancements, testing, and technical documentation.
- Collaborated on architecture discussions, feature development, debugging, deployment support, and engineering best practices to ensure scalability, maintainability, and reliability.
- Previously contributed to the Patent Analytics Tool (PAT), enhancing patent search, classification recommendations, reporting capabilities, and supporting application compliance, security assessments, and deployments.
- Authored architecture documentation, technical guides, and knowledge-transfer material to support onboarding and long-term maintainability.

Aug 2021 – April 2024

**TATA CONSULTANCY SERVICES, Bengaluru** – *System Engineer*

- Supported infrastructure monitoring and operations using SolarWinds and NewRelic, ensuring system stability and operational readiness.
- Played a key role in a data-driven transformation initiative for a leading retail client, addressing the challenge of analyzing 65,000+ monthly alerts to improve IT operational efficiency.
- Drove the adoption of Power BI for alert analytics and reporting, replacing manual workflows and reducing analysis time from hours to minutes.
- Built and maintained a centralized data repository integrated with Power BI, improving data accuracy and eliminating repetitive data-cleaning efforts.
- Developed interactive dashboards delivering actionable insights for stakeholders, including a comprehensive operational dashboard for Starbucks, supporting faster decision-making and improved system reliability.

## Projects

**CodeSmith_AI – Autonomous AI Developer** - [Python | CrewAI | OpenAI | Agentic AI | Docker | Prompt Engineering]

- Designed and developed an autonomous AI coding assistant that plans, generates, executes, and evaluates code from natural language instructions using multi-agent orchestration.
- Implemented CrewAI-based agent workflows to coordinate task understanding, code generation, execution, and result validation, producing structured, human-readable outputs.
- Integrated containerized code execution using Docker, enabling secure, isolated, and reproducible runtime environments for safe evaluation of generated programs.
- Built a JSON-based response framework returning generated code, execution results, logs, and metadata to support downstream automation and analysis.
- Enabled dynamic multi-language execution (currently optimized for Python) with safety controls such as runtime limits and automated retry mechanisms to handle failures gracefully.
- Enhanced developer experience by implementing a Rich-powered terminal UI with syntax highlighting and structured output visualization.

**Deep Research Analysis – AI-Powered Research Orchestrator** – [Python | OpenAI SDK | Agentic AI | Async Processing | Gradio | Prompt Engineering]

- Designed and developed an AI-driven research orchestration system that automates end-to-end topic exploration, content synthesis, and report generation using multiple asynchronous agents.
- Implemented an agent-based pipeline to perform web research, summarize findings, and synthesize insights into a structured Markdown report, reducing manual research effort significantly.
- Built an interactive Gradio interface enabling users to trigger deep research workflows and visualize intermediate and final outputs in a clean, user-friendly UI.
- Orchestrated asynchronous task execution to improve throughput and responsiveness while coordinating search, analysis, and report generation stages.
- Automated report delivery via email, enabling seamless sharing of finalized research outputs without manual intervention.

**StockPicker AI - Intelligent Investment Research Assistant** - [Python | Agentic AI | Multi-Agent Systems | OpenAI SDK | Async Workflows | Prompt Engineering]

- Designed and built an autonomous, multi-agent investment research system that identifies trending companies, performs deep financial analysis, and delivers personalized investment recommendations.
- Implemented intelligent company discovery pipelines to analyze market news, earnings reports, and sector trends while ensuring non-repetitive, unbiased selections.
- Developed deep financial analysis of workflows evaluating fundamentals, financial ratios, competitive positioning, management quality, growth potential, and risk factors.
- Engineered a deep financial analysis workflow combining quantitative and qualitative criteria, providing transparent rationale for top picks and exclusion decisions.
- Orchestrated 7 specialized AI agents with sequential task execution and context passing to ensure coherent analysis across discovery, evaluation, and recommendation stages.
- Automated multi-channel reporting, delivering executive-style email summaries and real-time mobile notifications via PushOver with consistent messaging.

## Certifications

- Claude Certified Developer Foundations (CCDV-F)
- Claude Certified Associate Foundations (CCAO-F)

## Education

Aug 2017 – July 2021

**Visvesvaraya Technological University, Belagavi** - *Bachelor of Engineering (Computer Science & Engineering)*